

Bootstrapping the Architecture 2.0 Ecosystem

PHASE 1: OPEN FOUNDATIONS

STEP 1

Open data schemas & tool contracts

- Open Agentic Interface (OAI)
- SLSA attempt lineage headers
- Protobuf / JSON-LD schemas
- Tool AI-readiness certs

STEP 2

Open PDKs & cloud silicon shuttles

- Sky130 & ASAP7 open PDKs
- Cloud MPW shuttle passes
- OpenROAD / OpenSTA flow
- AI shuttle telemetry grants

PHASE 2: GOVERNANCE

STEP 3

Consortium governance & matched benchmarks

- MLCommons / SPEC working group
- Reference problem suites:
3 W XR, RISC-V RV64GCV,
CXL & UCle chiplet targets
- Closed division rules:
locked wrappers + compute
- Open division rules:
unconstrained surrogates
- Total workflow cost accounting:
GPU hours + license minutes
plus human triage time

PHASE 3: ACCELERATION

STEP 4

Grand challenges & architecture contests

- ISCA / MICRO / DAC tracks
- Constrained PPA generation
- Sample-efficient DSE
- Red-teaming & SVA security

STEP 5

Public repositories of failed design attempts

- Unroutable floorplan logs
- Negative-slack STA reports
- UPF power domain conflicts
- Negative controls for physics

PHASE 4: SCALE & SIGN-OFF

STEP 6

Workforce curricula & silicon sign-off

- University curriculum pivot:
formulation + verification
- Diagnostic fault-injection labs
- Autonomous Workflow Certified
(AWC) conference badges
- Supported architecture claims
for tapeout commitment
- Living hub: arch2.mlsysbook.ai
schemas + docker starter kits
- Democratized ASIC economics

Negative controls calibrate surrogate physical feasibility

Foundational data schemas and physical signoff harnesses must precede unconstrained generation or search.